1. Reinforcement Learning (RL)

- **Definition:** A computational approach to learning where an **agent** learns to achieve a goal in a complex, uncertain **environment** by maximizing a numerical **reward** signal.
- **The RL Loop:** * The agent perceives the **State** of the environment.
    - It performs an **Action** based on its current **Policy**.
    - The environment provides a **Reward** (positive or negative) and transitions to a new state.
- **Exploration vs. Exploitation:** A core challenge where the agent must choose between trying new actions to find better rewards (exploration) and using known actions that yield high rewards (exploitation).
- **Example (Robot Navigation):** A robot learning to navigate a maze. Hitting a wall results in a negative reward (-10), while reaching the exit provides a large positive reward (+100). The robot eventually learns the shortest path to maximize its score.

---

2. Partial Order Planning (POP)

- **Concept:** A planning paradigm that represents plans as a set of actions with temporal constraints, rather than a rigid linear sequence.
- **Least Commitment Principle:** Decisions about the order of actions are delayed as long as possible. If two actions do not depend on each other, they remain unordered.
- **Components:**
    - **Actions:** Steps with defined preconditions and effects.
    - **Ordering Constraints:** $A < B$ means Action A must occur before Action B.
    - **Causal Links:** Used to track which action satisfies a specific precondition for a subsequent action.
- **Example:** In a "Fixing a Flat Tire" plan, the actions "Remove spare from trunk" and "Loosen lug nuts" are independent and can be performed in any order, whereas "Jack up car" must happen before "Remove flat tire."

---

3. Hill Climbing Algorithm

- **Definition:** An iterative local search algorithm that moves continuously in the direction of increasing value—essentially an optimization technique for finding the "peak" of a mathematical function.
- **Process:** It starts with a random solution and examines its immediate neighbors. If a neighbor provides a better value, it becomes the new current state. This repeats until no neighbor is better.
- **Drawbacks:**

- **Local Maxima:** The algorithm stops at a peak that is higher than its neighbors but lower than the global optimum.
- **Ridges:** Slopes that lead to a sequence of local maxima, making it difficult for the algorithm to navigate.
- **Plateaus:** Flat areas where the evaluation function returns the same value for all neighbors, causing the search to wander aimlessly.

- **Example:** Maximizing a chemical yield by slightly adjusting temperature and pressure. You keep increasing them until the yield starts to drop, at which point you stop.

---

4. Alpha-Beta Pruning

- **Purpose:** An adversarial search optimization that reduces the number of nodes evaluated by the **Minimax algorithm** in game trees.
- **The Variables:**
  - **Alpha (α):** The best value that the **Maximizer** can currently guarantee at that level or above.
  - **Beta (β):** The best value that the **Minimizer** can currently guarantee at that level or above.
- **The Mechanism:** If at any point $\alpha \geq \beta$, the Maximizer or Minimizer knows that the current branch will never be chosen by the opponent, so the rest of that branch is "pruned" (ignored).
- **Significance:** It allows the AI to search twice as deep in the same amount of time, which is critical for high-level play in games like Chess or Go.

---

5. Local Search Algorithms

- **Core Philosophy:** Unlike systematic searches (like BFS/DFS), local search moves from one state to a neighboring state without keeping track of the path. They are memory-efficient.
- **Key Algorithms:**
  - **Simulated Annealing:** Mimics the cooling of metal. It allows "downhill" (bad) moves early on to escape local maxima, with the probability of such moves decreasing over "time" (temperature).
  - **Genetic Algorithms:** Uses a population of solutions. It employs **Selection** (choosing the best), **Crossover** (combining two solutions), and **Mutation** (randomly changing a solution) to evolve better results.
- **Example:** The **Traveling Salesperson Problem (TSP)**, where local search tries swapping the order of cities to find the shortest total route.

---

6. PAC Learning Model

- **Full Name:** Probably Approximately Correct learning.
- **Mathematical Goal:** To provide a formal framework for what it means for a machine to "learn" a concept from data.
- **Two Parameters:**
    - **Error (ϵ):** The hypothesis is "approximately correct" if its error rate is less than ϵ.
    - **Confidence (δ):** The learner will "probably" succeed if it finds such a hypothesis with a probability of at least $1 - \delta$.
- **Significance:** It helps determine the **sample complexity**—how many training examples are required to ensure the model learns correctly within a specific margin of error.

---

7. Belief Networks (Bayesian Networks)

- **Definition:** A probabilistic graphical model that uses a **Directed Acyclic Graph (DAG)** to represent variables and their conditional dependencies.
- **Structure:**
    - **Nodes:** Represent random variables (e.g., "Rain," "Sprinkler," "Grass Wet").
    - **Edges:** Represent direct influence or causal relationships.
- **Conditional Probability Tables (CPT):** Each node contains a table showing the probability of its states given the states of its "parent" nodes.
- **Example:** Detecting a medical condition. "High Blood Pressure" might be a node influenced by "Diet" and "Genetics." The network calculates the probability of the condition based on which factors are present.

---

8. Basics of Natural Language Processing (NLP)

- **Definition:** The intersection of AI and linguistics aimed at enabling computers to understand, interpret, and generate human language.
- **Fundamental Levels of Analysis:**
    - **Phonological:** Analyzing the sounds of speech.
    - **Morphological:** Analyzing the structure of words (roots, prefixes, suffixes).
    - **Lexical:** Identifying and analyzing individual words (Tokenization).
    - **Syntactic:** Analyzing the grammatical structure of sentences (Parsing).
    - **Semantic:** Determining the literal meaning of words and sentences.
    - **Pragmatic:** Understanding how context influences meaning (e.g., sarcasm or metaphors).
- **Applications:** Machine translation (Google Translate), Sentiment Analysis (reading reviews), and Virtual Assistants (Siri/Alexa).